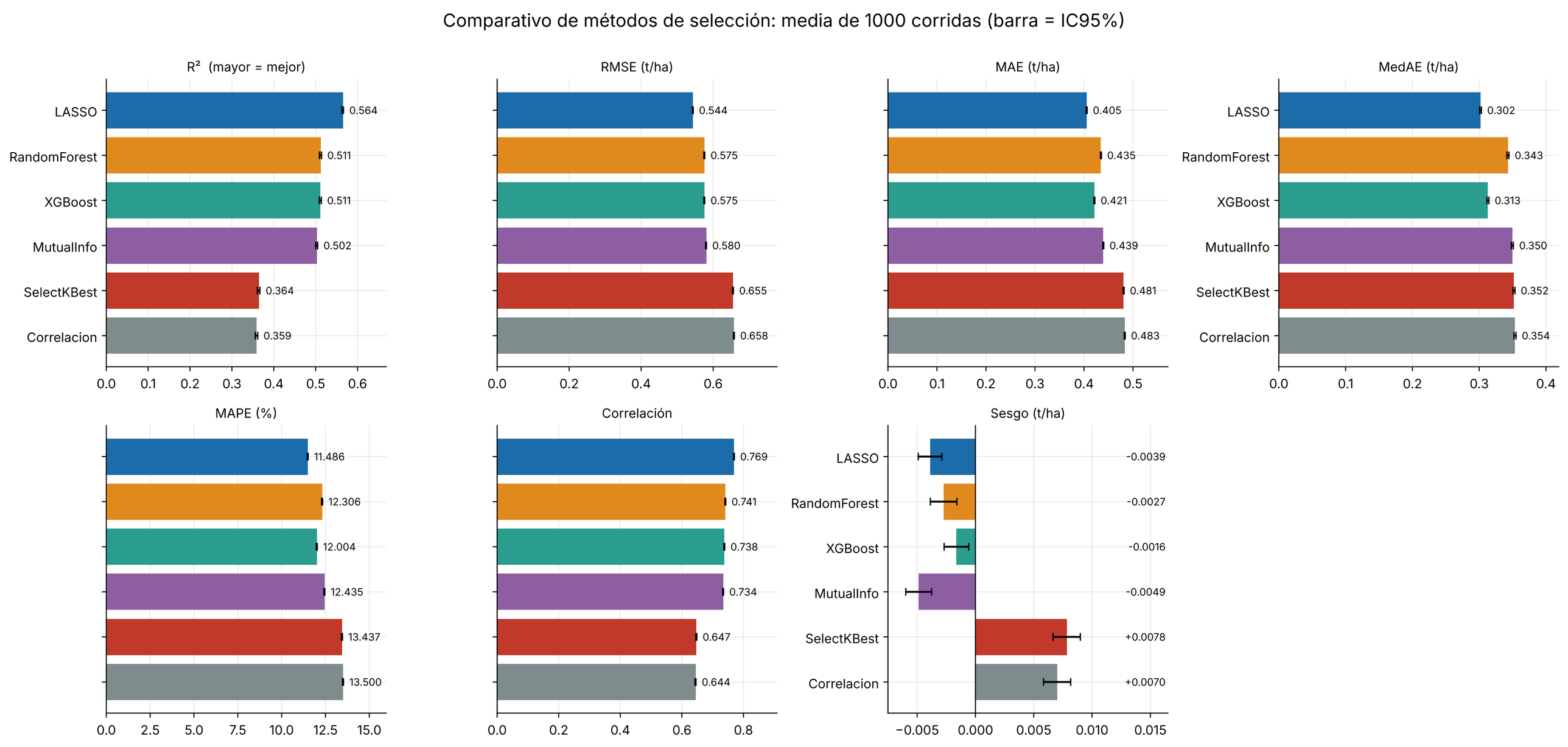
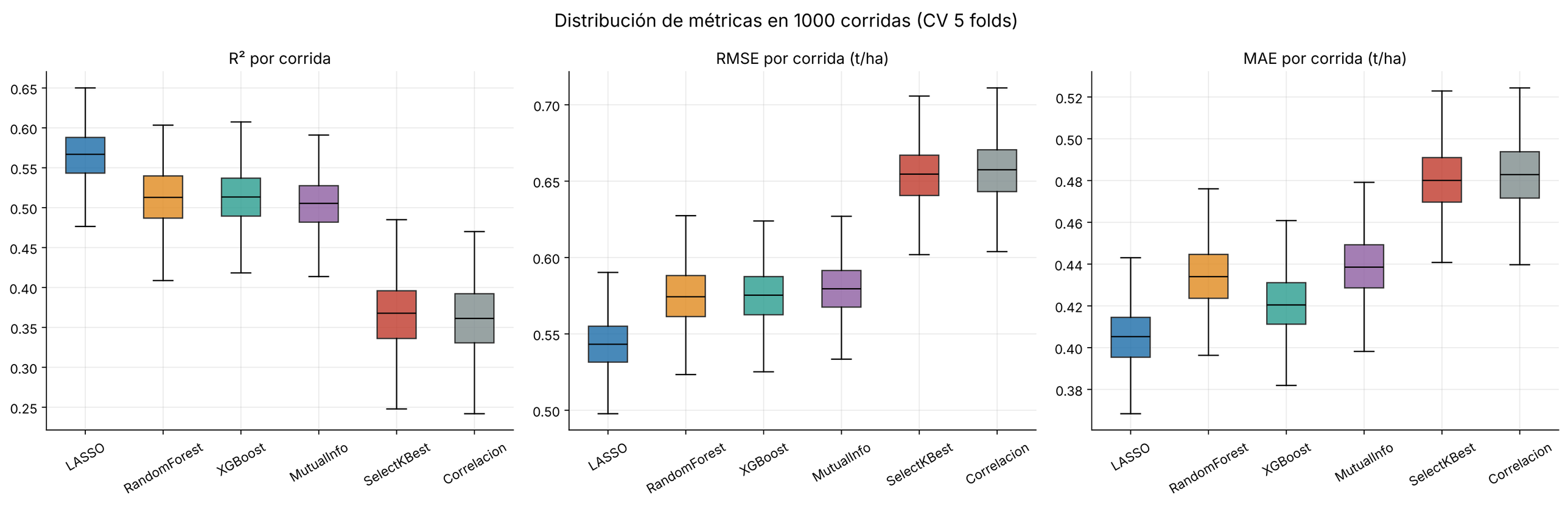
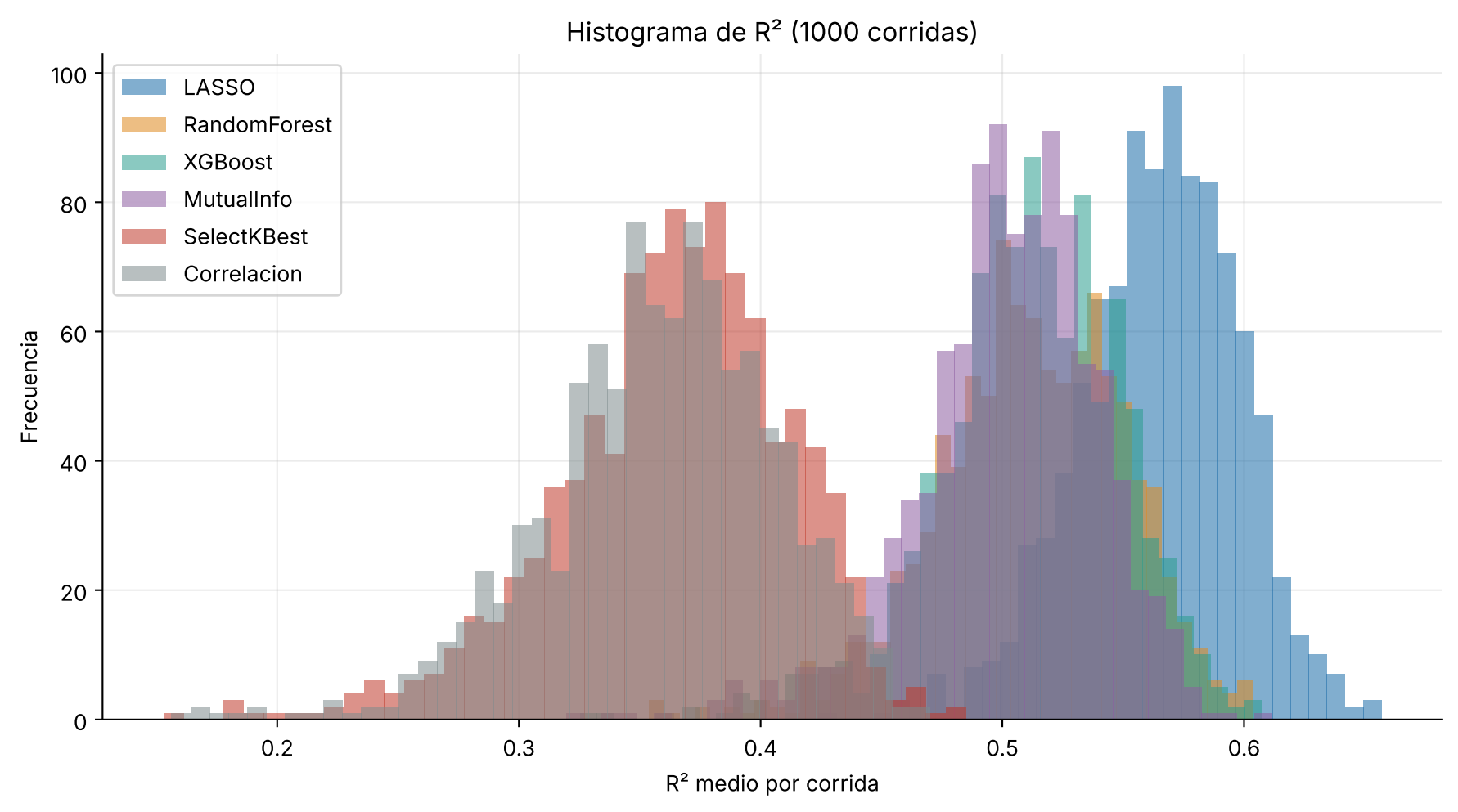
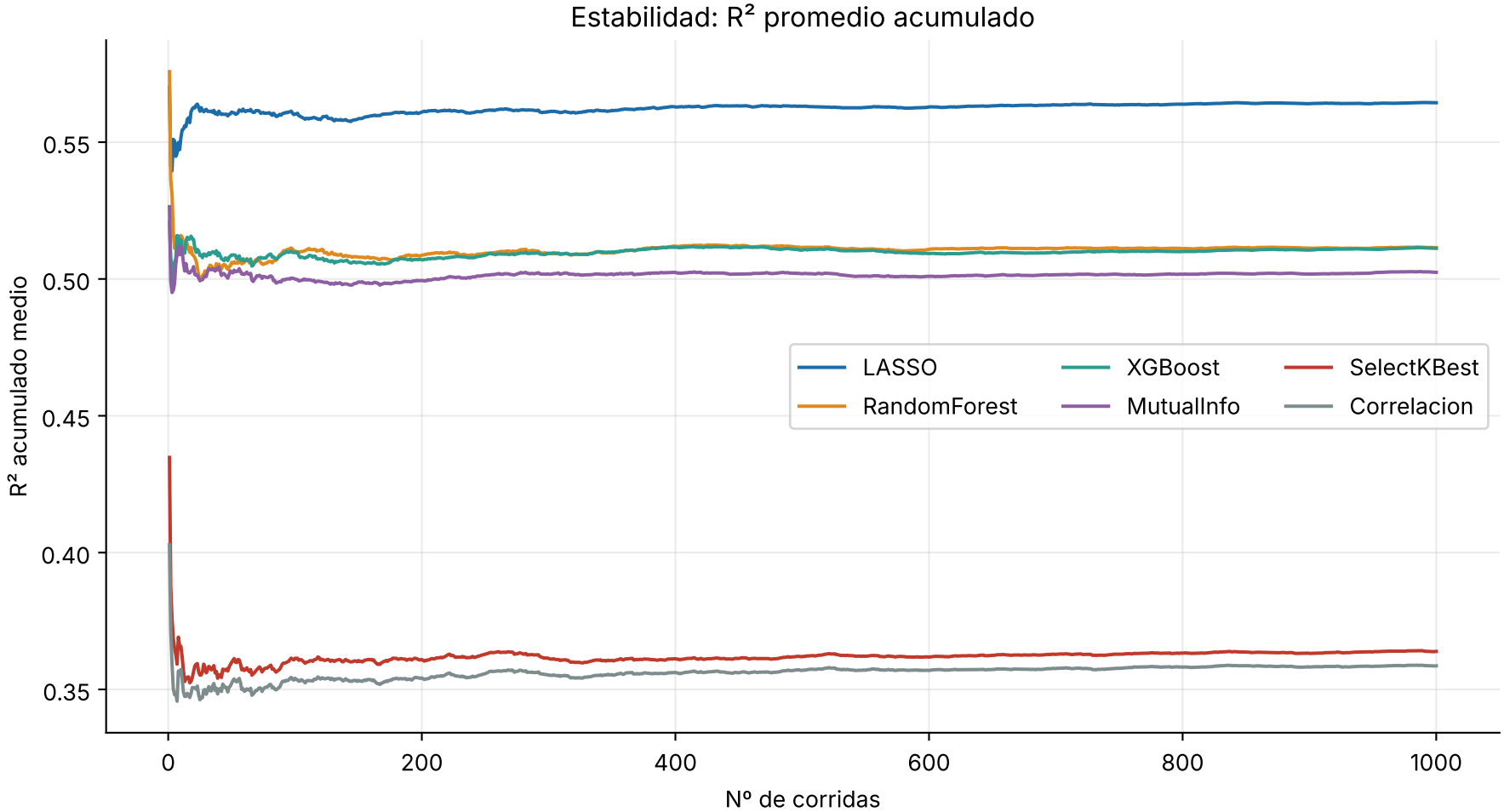
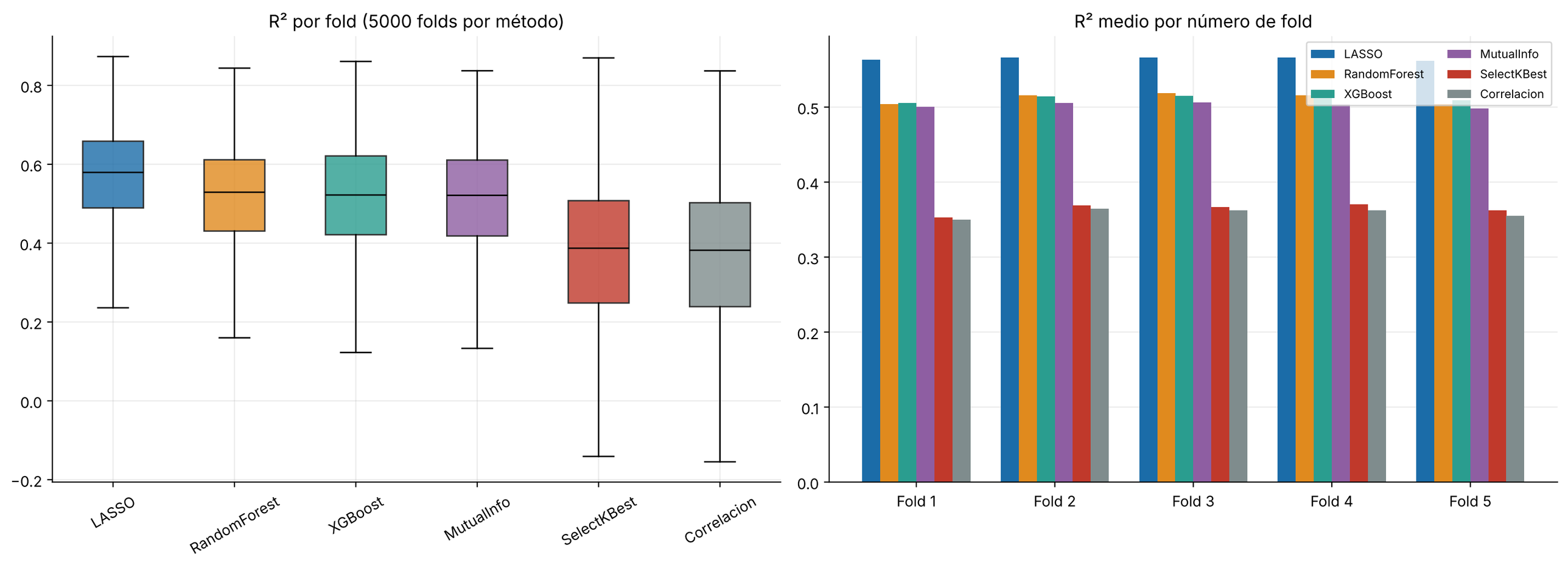
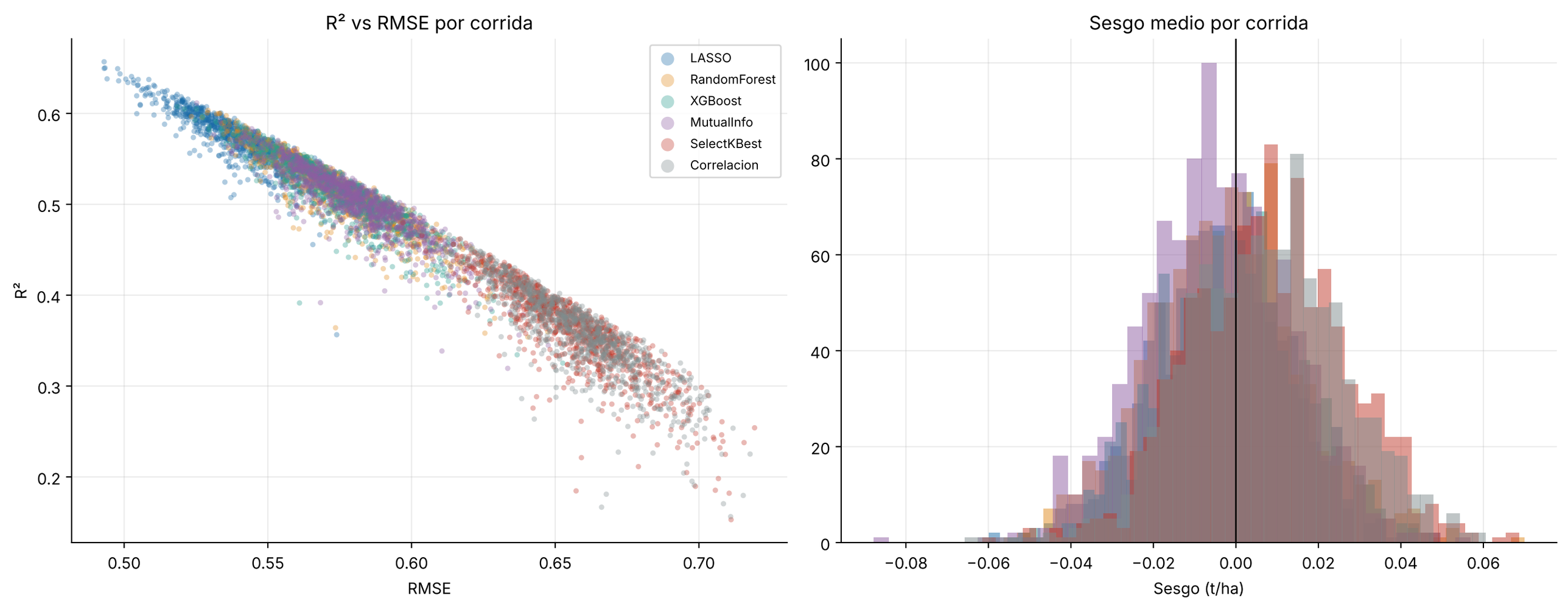

Comparativo de métodos de selección: media de 1000 corridas (barra = IC95%)
R² (mayor = mejor)
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion
0.564
0.511
0.511
0.502
0.364
0.359
0.0 0.1 0.2 0.3 0.4 0.5 0.6
RMSE (t/ha)
0.544
0.575
0.575
0.580
0.655
0.658
0.0 0.2 0.4 0.6
MAE (t/ha)
0.405
0.435
0.421
0.439
0.481
0.483
0.0 0.1 0.2 0.3 0.4 0.5
MedAE (t/ha)
0.302
0.343
0.313
0.350
0.352
0.354
0.0 0.1 0.2 0.3 0.4
MAPE (%)
11.486
12.306
12.004
12.435
13.437
13.500
0.0 2.5 5.0 7.5 10.0 12.5 15.0
Correlación
0.769
0.741
0.738
0.734
0.647
0.644
0.0 0.2 0.4 0.6 0.8
Sesgo (t/ha)
-0.0039
-0.0027
-0.0016
-0.0049
+0.0078
+0.0070
−0.005 0.000 0.005 0.010 0.015

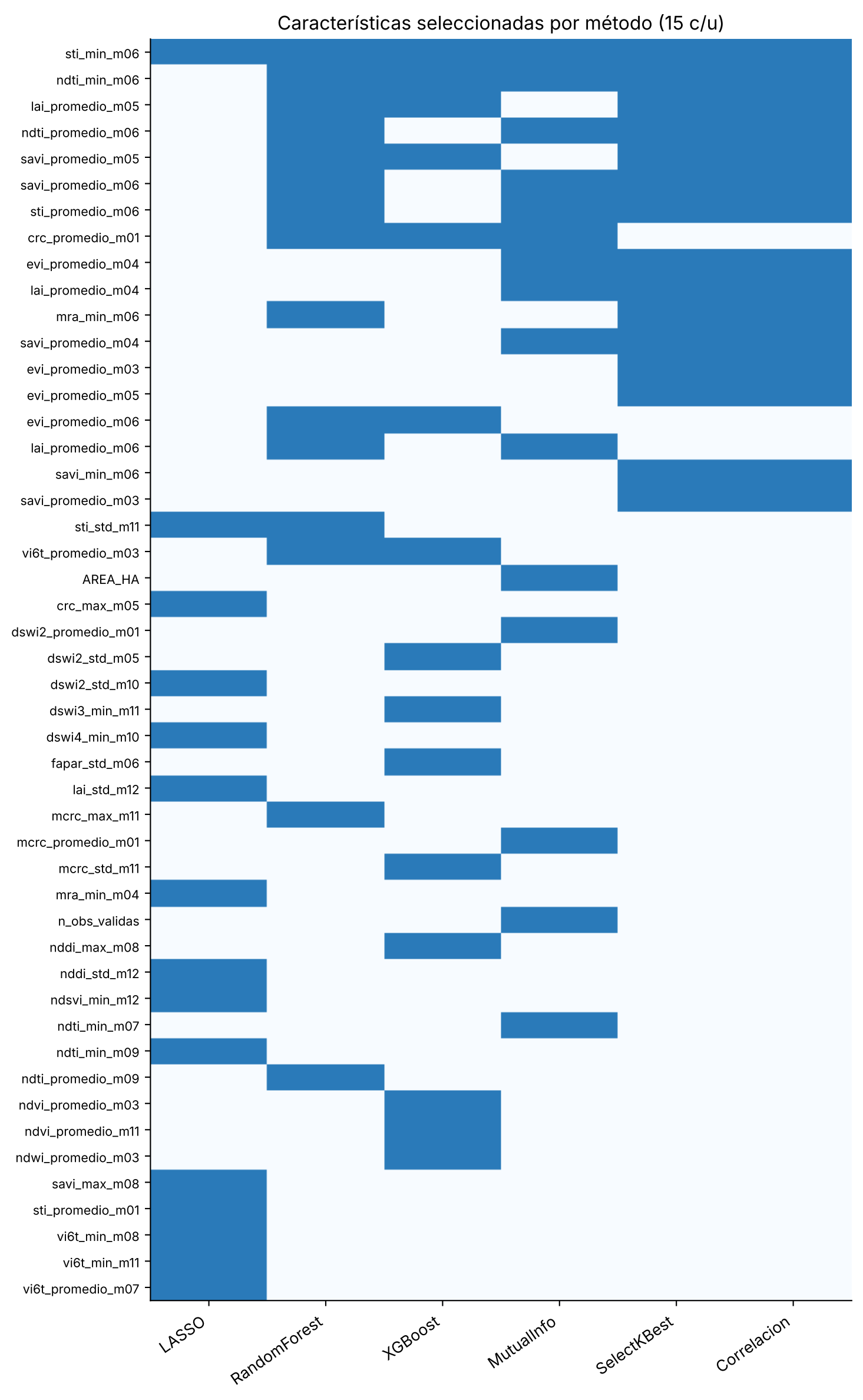

Distribución de métricas en 1000 corridas (CV 5 folds)
R² por corrida
RMSE por corrida (t/ha)
MAE por corrida (t/ha)
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion

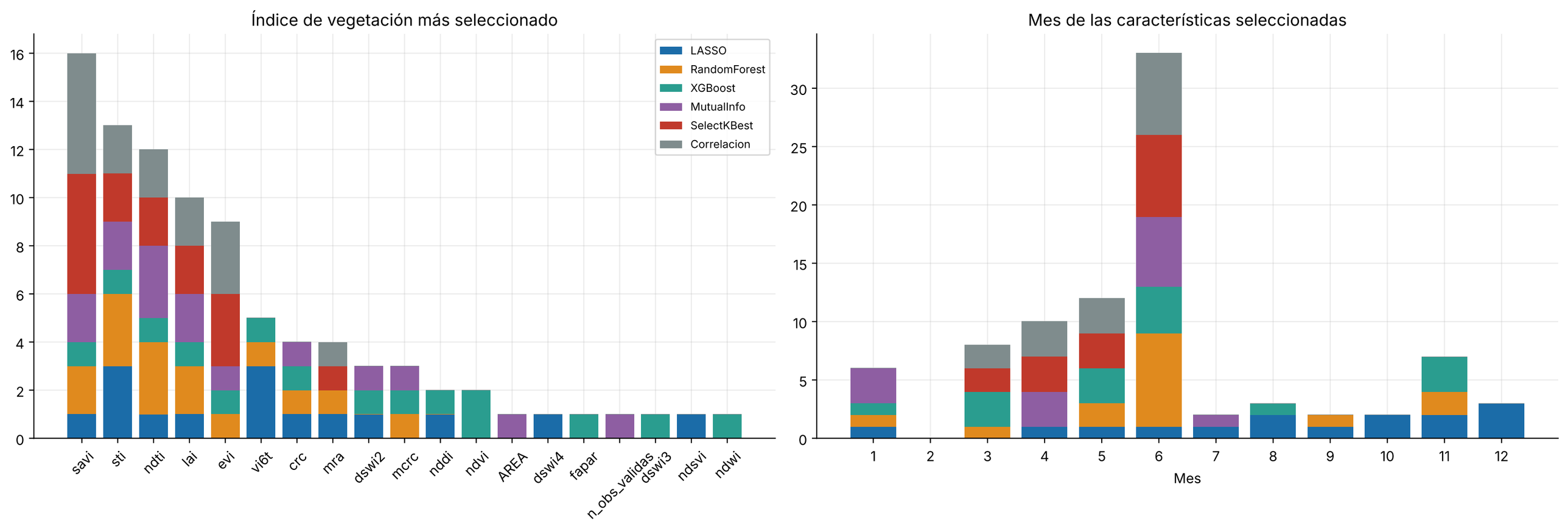

Histograma de R² (1000 corridas)
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion
Frecuencia
R² medio por corrida
0
20
40
60
80
100
0.2
0.3
0.4
0.5
0.6

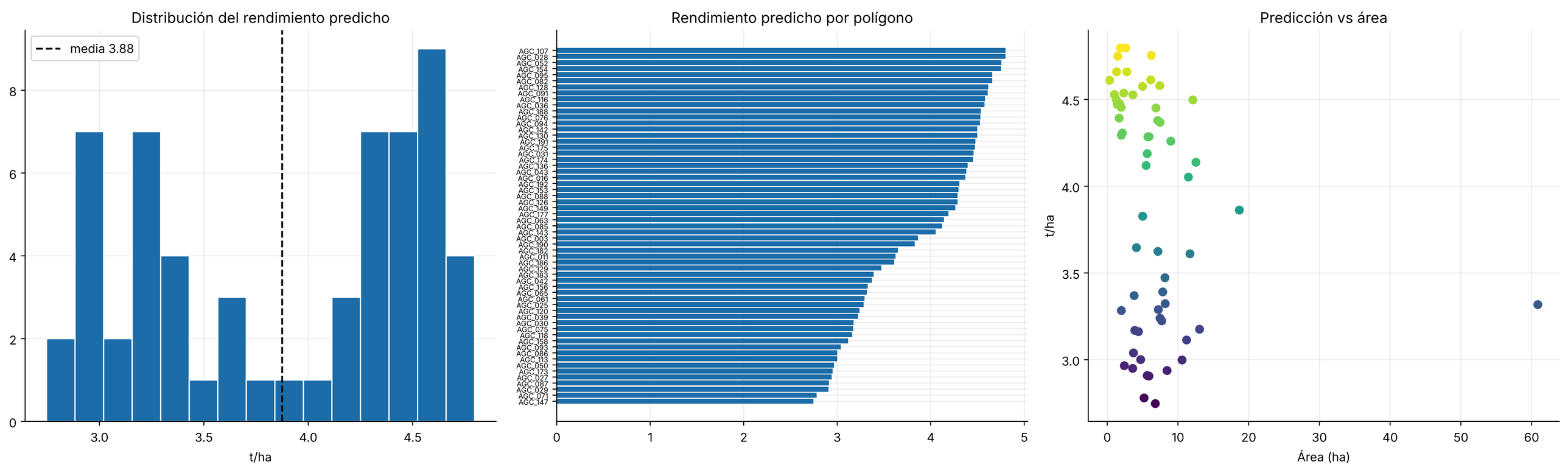

Estabilidad: R² promedio acumulado
R² acumulado medio
0.55
0.50
0.45
0.40
0.35
0
200
400
600
800
1000
Nº de corridas
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion


R² por fold (5000 folds por método)
0.8
0.6
0.4
0.2
0.0
−0.2
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion
R² medio por número de fold
0.5
0.4
0.3
0.2
0.1
0.0
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion


R² vs RMSE por corrida
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion
R²
0.6
0.5
0.4
0.3
0.2
0.50
0.55
0.60
0.65
0.70
RMSE
Sesgo medio por corrida
100
80
60
40
20
0
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
Sesgo (t/ha)


Características seleccionadas por método (15 c/u)
sti_min_m06
ndti_min_m06
lai_promedio_m05
ndti_promedio_m06
savi_promedio_m05
savi_promedio_m06
sti_promedio_m06
crc_promedio_m01
evi_promedio_m04
lai_promedio_m04
mra_min_m06
savi_promedio_m04
evi_promedio_m03
evi_promedio_m05
evi_promedio_m06
lai_promedio_m06
savi_min_m06
savi_promedio_m03
sti_std_m11
vi6t_promedio_m03
AREA_HA
crc_max_m05
dswi2_promedio_m01
dswi2_std_m05
dswi2_std_m10
dswi3_min_m11
dswi4_min_m10
fapar_std_m06
lai_std_m12
mcrc_max_m11
mcrc_promedio_m01
mcrc_std_m11
mra_min_m04
n_obs_validas
nddi_max_m08
nddi_std_m12
ndsvi_min_m12
ndti_min_m07
ndti_min_m09
ndti_promedio_m09
ndvi_promedio_m03
ndvi_promedio_m11
ndwi_promedio_m03
savi_max_m08
sti_promedio_m01
vi6t_min_m08
vi6t_min_m11
vi6t_promedio_m07
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion


Índice de vegetación más seleccionado
LASSO
RandomForest
XGBoost
MutualInfo
SelectKBest
Correlacion
0
2
4
6
8
10
12
14
16
savi
sti
ndti
lai
evi
vi6t
crc
mra
dswi2
mcrc
nddi
ndvi
AREA
dswi4
fapar
n_obs_validas
dswi3
ndsvi
ndwi
Mes de las características seleccionadas
0
5
10
15
20
25
30
1
2
3
4
5
6
7
8
9
10
11
12
Mes


Distribución del rendimiento predicho
media 3.88
t/ha
Rendimiento predicho por polígono
AGC_107
AGC_028
AGC_052
AGC_154
AGC_095
AGC_082
AGC_128
AGC_091
AGC_116
AGC_036
AGC_188
AGC_076
AGC_094
AGC_142
AGC_130
AGC_191
AGC_175
AGC_031
AGC_174
AGC_136
AGC_043
AGC_016
AGC_192
AGC_153
AGC_088
AGC_125
AGC_149
AGC_177
AGC_063
AGC_085
AGC_143
AGC_003
AGC_190
AGC_182
AGC_011
AGC_186
AGC_129
AGC_183
AGC_042
AGC_156
AGC_065
AGC_061
AGC_025
AGC_120
AGC_039
AGC_030
AGC_075
AGC_118
AGC_158
AGC_093
AGC_086
AGC_113
AGC_050
AGC_173
AGC_027
AGC_087
AGC_029
AGC_071
AGC_147
Predicción vs área
t/ha
Área (ha)